

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 12

BIOTECHNOLOGY - 2

BIOTECHNOLOGY

4 INTERFERON PRODUCTION

Interferons are signaling proteins (Cytokines) that have the ability to stop viruses.

- When there is a viral invasion, the viral infected cell releases interferons causing other cells to increase their anti-viral defences (it causes the synthesis of several enzymes which inhibit viral growth).
- The interferons also activate White Blood Cells which attack and destroy the viruses.
- Interferons were named for its ability to interfere with viral growth in the body.
- Interferons also fight against bacterial and parasitic infections.
- Interferons are used as a treatment for common cold, influenza, hepatitis, herpes etc.
- In 1980, American scientists successfully produced interferons in lab using Gene Cloning in bacteria which is found in the intestines.
- Later Human Leukocyte Interferon was produced by attaching a gene sequence to yeast. The yeast cells could produce about a million molecules of interferon per cell at a very low cost.

APPLICATIONS OF BIOTECHNOLOGY

5 FERMENTATION BIOTECHNOLOGY

Fermentation Definition

- a) It is the technique by which useful organic compounds are prepared from less useful ones by the use of enzymes or cells.
- b) Any process for the production of a product by means of mass culture of micro-organisms.



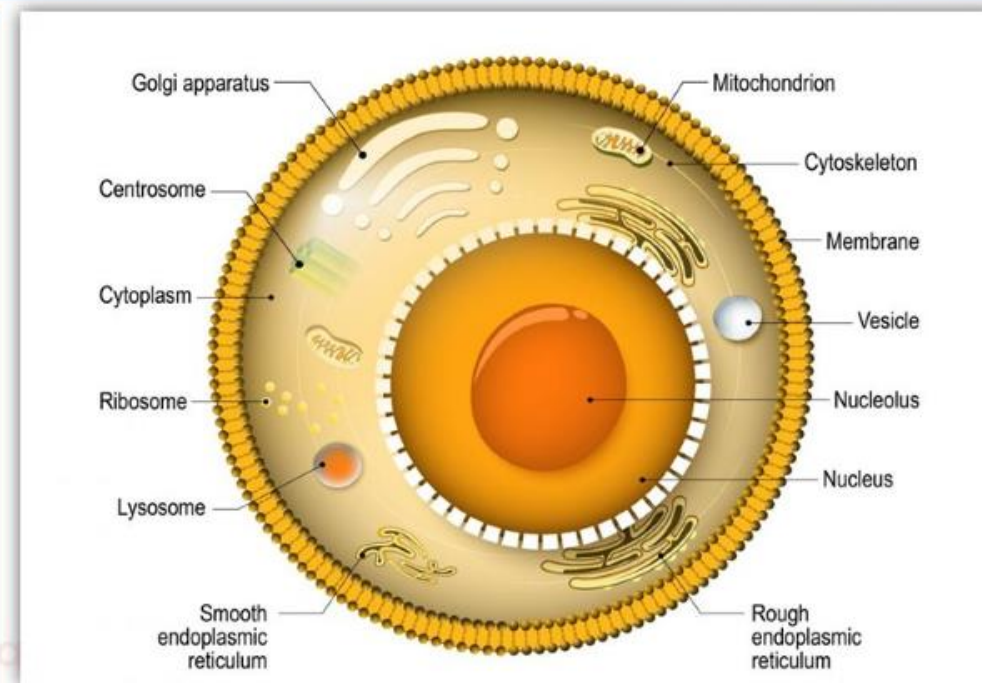
5 FERMENTATION BIOTECHNOLOGY

- Fermentation technology aims at making this process more efficient, productive and economical.
- Fermentation technology is the oldest of all biotechnological processes.
- Ancient fermented food processes, such as making bread, wine, cheese, curds, idli, dosa, etc., some of which are 6,000 yr old and developed long before man had any knowledge of the existence of the micro-organisms involved, also genuinely constitute biotechnology.
- Fermentation industry uses agricultural products as raw materials.
- This capability of living cells is used in fermentation.
- The juice of grapes containing glucose is broken down into ethyl alcohol by yeast cells.



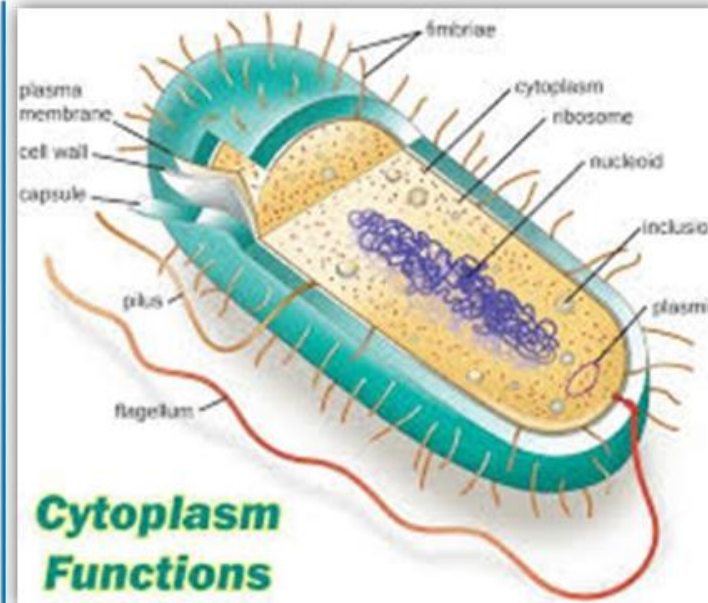
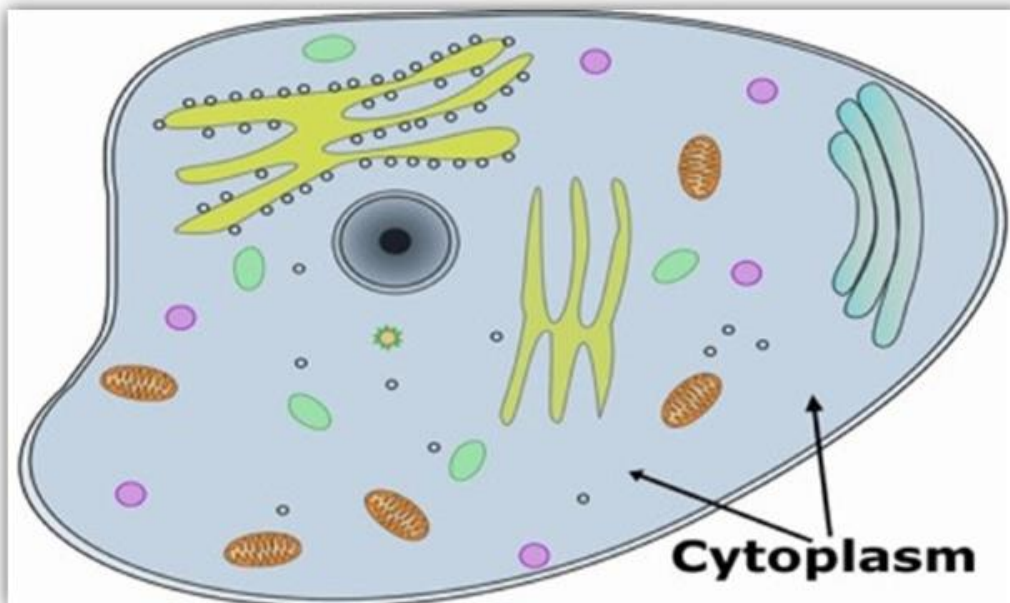
BASIC CONCEPTS

- Cell was discovered in 1665 by Robert Hooke who also coined the term.
 - Robert Brown discovered the cell nucleus in 1831.
 - Schleiden and Schwann formulated the Cell Theory (1839) which states that all living organisms are composed of cells and cell is the basic unit of life.
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- With the invention of electron microscope in 1940, it was possible to observe and understand the complex structure of cell.



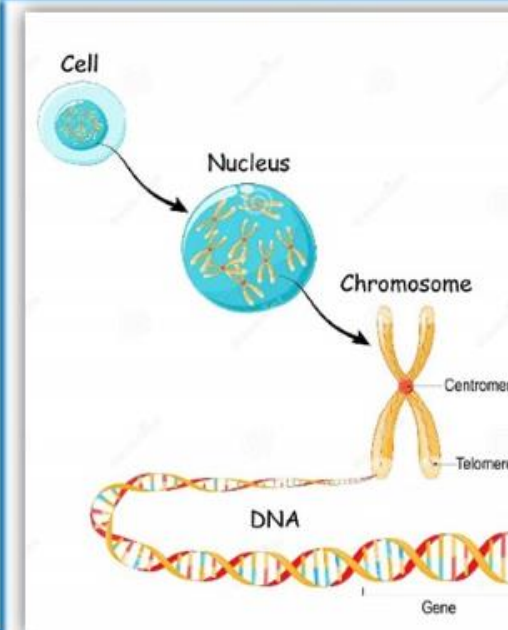
CYTOPLASM

- It is the substance present between the cell membrane (outermost covering of cell) and the nucleus.
- It is a fluid like substance which contains cell organelles – Mitochondria, Golgi complex, Ribosomes etc.
- Each of these organelles perform a specific function.



NUCLEUS

- It is spherical and is located in the center of the cell.
 - It is separated from the cytoplasm by a double layered membrane called Nuclear Membrane.
 - The membrane is porous and allows the movement of materials between the cytoplasm and the nucleus.
 - Nucleus acts as a control center of the activities of cell
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- Cytoplasm + Nucleus is Protoplasm, which is the living substance of the cell.
 - Nucleus contains thread like structures called Chromosomes.
 - These carry genes and help in inheritance or transfer of characteristics from the parents to the offspring.
 - Gene is the unit of inheritance in living organisms. It controls the transfer of hereditary characteristics from parents to offspring.



NUCLEUS

- Genes carry the information that determines your traits, which are features or characteristics that are passed on to you (inherited) from your parents.
- Genes are made up of DNA. Some genes act as instructions to make molecules called proteins.
- So, chromosomes contain genes which contain information for inheritance of features from parents to offspring in the form of DNA.
- DNA molecules contain the information necessary for constructing and organizing cells.
- Chromosomes can be seen only when the cell divides. In a cell which is not dividing, the DNA is present as part of chromatin material.
- Chromatin is visible as an entangled mass of thread like structures. Whenever the cell is about to divide the chromatin material gets separated into chromosomes.

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